Luca **Corbucci**

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About Me

Postdoctoral Researcher in Multimodal LLMs at FBK (Trento, Italy). I hold a PhD in Computer Science focused on AI Safety for Federated Learning. Authored 5+ publications in top-tier venues, including **FAccT** and **ECAI**. Proven leader and collaborator: successfully initiated a 1-year research internship with **Telefónica Research Lab** and projects with **Johannes Gutenberg-Universität Mainz**. Experienced public speaker, community builder (FOSDEM AI Devroom, Superhero Valley, Pisa.dev), and **podcaster** (280+ episodes).

Education

University of Pisa - KDD Lab

Pisa, Italy

PhD in Computer Science - **Advisors:** Prof. A. Monreale and Dr. N. Kourtellis. - **Final Grade: Summa Cum Laude** *Nov. 2021 - Jul. 2025*

- Developed methodologies to ensure privacy and fairness in Federated Learning models, addressing key challenges in **safe and ethical AI**.
- Explored and designed techniques for private and explainable Federated Learning (**XAI** for FL) to enhance model transparency and trust.
- Started collaboration with **Telefonica Research Lab** (1-year visit) and **Johannes Gutenberg-Universität Mainz**.
- Partially funded and supported by the Horizon Europe project "TANGO".
- Supervised **3 master's theses**, including one that led to a peer-reviewed publication at ECAI 2025.

Technical University of Munich

Munich, Germany

Computer Science - Erasmus+ Study *Oct. 2019 - Mar. 2020*

- Won a 5-month Erasmus+ Study grant to visit the Technical University of Munich.
- **Courses:** Machine Learning, Basic Mathematical Tools for Imaging and Visualization, Practical Course: Cloud Databases.

University of Pisa

Pisa, Italy

M.Sc. in Computer Science - **Final Grade:** 110/110 Cum Laude *Oct. 2018 - Jun 2021*

- **Thesis:** "Semantic enrichment of XAI explanations for healthcare". Paper published at the Discovery Science Conference 2023. [**Paper** | **Code**]
- Supervised by: Prof. Anna Monreale and Dr. Cecilia Panigutti.
- Spent a semester at **Technical University of Munich** thanks to the Erasmus+ study program.

University of Pisa

Pisa, Italy

B.Sc. in Computer Science *Sep. 2014 - Oct. 2018*

- **Thesis:** "Construction and Analysis of the Ethereum Transaction Graphs". [**Code**]

Work Experience

Telefónica R&D - System AI Lab

Barcelona, Spain

Research Intern - Supervised by Dr. Nicolas Kourtellis *Mar. 2023 - Mar. 2024*

- Designed PUFFLE, a **Federated Learning** method achieving 75% fairness improvement with minimal (22%) accuracy loss, while preserving model privacy and utility. [**Paper** | **Code**]
- Integrated fairness constraints based on practitioner requirements, bridging the gap between theoretical fairness and real-world systems.
- **Paper** accepted at the European Conference on Artificial Intelligence (**ECAI**) 2024; **Patent** application submitted.
- Partially funded by the SPATIAL H2020 EU project.

University of Pisa - Computer Science Department

Pisa, Italy

Teaching Assistant for the "Programming and Algorithmics" Bachelor Course

- Provided support to students over three semesters, helping them with algorithms, data structures, and programming.
- Organized exercise sessions to enhance students' problem-solving skills and algorithmic thinking.

University of Pisa

Pisa, Italy

Research Assistant *Jun. 2021 - Oct. 2021*

- Won a research scholarship to work on the semantic enrichment of Explainable AI methods for healthcare.
- The results of this work have been published in a paper presented at Discovery Science Conference 2023.

Tocket

Pisa, Italy

Flutter Developer - Tech Stack: Flutter, Dart, Python, Flask *Mar. 2020 - Jan. 2021*

- Developed the Tocket mobile app released on the iOS App Store and Google Play Store to buy tickets for online and in-person events.
- More than **20k users** have downloaded and used the app.
- Developed internal APIs and dashboard.

Academic Roles

- Reviewer for European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases **ECML PKDD 2024 and 2025**
- Reviewer for European Conference on Artificial Intelligence **ECAI 2023**

Publications

Under Review	**L. Corbucci**, M. Heikkilä, D. Noguero, A. Monreale, N. Kourtellis, *"Self-Adaptive Balancing of Fairness, Privacy, and Utility in Federated Learning"*	
Under Review	V. Bonsignori, **L. Corbucci**, F. Naretto, A. Monreale, "*Interpreting Federated Learning by Aggregating SHAP Explanations*"	
Under Review	X. Heilmann, **L. Corbucci**, M. Cerrato, Anna Monreale, *"FeDa4Fair: Client-Level Federated Datasets for Fairness Evaluation"*	
Oct 2025	**L. Corbucci**, J. A. B. Legrottaglie, F. Spinnato, A. Monreale and R. Guidotti, "*An Interpretable Data-Driven Unsupervised Approach for the Prevention of Forgotten Items*" - Accepted at European Conference on Artificial Intelligence (ECAI) 2025	*Bologna, Italy*
Sep 2025	**L. Corbucci**, F. Naretto, A. Monreale, "*Enhancing Local Explanations with GAN-Based neighborhood generation*" - Accepted at International Conference on Discovery Science 2025	*Ljubljana, Slovenia*
Jun 2025	X. Heilmann, **L. Corbucci**, M. Cerrato, "*A Benchmark for Client-level Fairness in Federated Learning*" - Accepted at European Workshop on Algorithmic Fairness 2025	*Eindhoven, The Netherlands*
Jun 2025	**L. Corbucci**, X. Heilmann, M. Cerrato, *"Benefits of the Federation? Analyzing the Impact of Fair Federated Learning at the Client Level" - Accepted at ACM Conference on Fairness, Accountability, and Transparency (FAccT) 2025*	*Athens, Greece*
Jun 2025	V. Bonsignori, **L. Corbucci**, F. Naretto, A. Monreale, "*Differentially Private FastSHAP for Federated Learning Model Explainability*" - Accepted at International Joint Conference on Neural Networks (IJCNN) 2025	*Rome, Italy*
Oct 2024	**L. Corbucci**, M. Heikkilä, D. Noguero, A. Monreale, N. Kourtellis, *"PUFFLE: Balancing Privacy, Utility, and Fairness in Federated Learning"* - Accepted at European Conference on Artificial Intelligence (ECAI) 2024	*Santiago de Compostela, Spain*
July 2024	**L. Corbucci**, A. Monreale, R. Pellungrini, *"Enhancing Privacy and Utility in Federated Learning: A Hybrid P2P and Server-Based Approach with Differential Privacy Protection"* - Accepted at SECRYPT 2024	*Dijon, France*
Oct 2023	**L. Corbucci**, A. Monreale, C. Panigutti, M. Natilli, S. Smiraglio, D. Pedreschi, *"Semantic enrichment of explanations of AI models for healthcare"* - Accepted at International Conference on Discovery Science 2023	*Porto, Portugal*
Jul 2023	**L. Corbucci**, R. Guidotti, A. Monreale, *"Explaining Black-Boxes in Federated Learning"* - Accepted at World Conference on Explainable Artificial Intelligence	*Lisbon, Portugal*

Software Projects

PUFFLE (Written in Python, check on Github)

Pisa, Italy

A solution to balance Privacy, Utility, and Fairness in Federated Learning — *Mar. 2023 - Ongoing*

- Designed and developed a regularization-based method with Telefónica Research to ensure fairness and privacy in Federated Learning models.
- Enabled practitioners to define custom fairness requirements, bridging the gap between AI safety theory and real-world application.
- Achieved a 75% unfairness reduction with a minimal 22% accuracy trade-off.

Fire360 (Written in Python, check on Github)

Pisa, Italy

A comprehensive and fast local explanation approach tailored for tabular data. — *Feb. 2025 - Mar. 2025*

- Developed a computationally efficient local explainer that avoids costly per-instance generation by using a single synthetic neighborhood.
- Outperformed competing methods by 95% in explanation stability and robustness on multiclass datasets.

FastShap++ (Written in Python, check on Github)

Pisa, Italy

A differentially private and fast Federated Learning explainer — *Oct. 2024 - Feb. 2025*

- Developed an extension of the FastShap explainer for Federated Learning, integrating Differential Privacy to protect client data during the explanation process.

Leadership & Community Involvement

Devroom Manager

Brussels, Belgium

"Low-Level AI Engineering & Hacking" Dev Room @ Fosdem 2025 — *Feb 2025*

- Co-organized the "Low-Level AI Engineering & Hacking" Dev Room at Fosdem 2025, managing speaker selection, scheduling, and logistics.

Superhero Valley

Pisa, Italy

Co-founder and Community Manager — *Oct. 2020 - Ongoing*

- Co-founded and grew a community for 2000+ computer scientists, leading to 100+ members securing roles in Big Tech.
- Lead a team of **5** helpers in organising community activities and events.

PointerPodcast

Pisa, Italy

Co-Creator and Podcaster — *May. 2019 - Ongoing*

- Co-created a tech podcast with 250+ episodes and over **40k+** listens, featuring **70+** guests from academia and industry.

Pisa.dev *Pisa, Italy*

Co-founder and core organizer *Jul. 2022 - Ongoing*

- Co-founded Pisa.dev, a community for developers based in Pisa. Organized **20+ meetups and social events**.